

How category learning affects object representations: Not all morphspaces stretch alike

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ABSTRACT

How does learning to categorize objects affect how we visually perceive them?

Behavioral, neurophysiological, and neuroimaging studies have tested the degree to which category learning influences object representations, with conflicting results. Some studies find that objects become more visually discriminable along dimensions relevant to previously learned categories, while others find no such effect. One critical factor we explore here lies in the structure of the morphspaces used in different studies. Studies finding no increase in discriminability often use “blended” morphspaces, with morphparents lying at corners of the space. By contrast, studies finding increases in discriminability use “factorial” morphspaces, defined by separate morphlines forming axes of the space. Using the same four morphparents, we created both factorial and blended morphspaces matched in pairwise discriminability. Category learning caused a selective increase in discriminability along the relevant dimension of the factorial space, but not in the blended space, and led to the creation of functional dimensions in the factorial space, but not in the blended space. These findings demonstrate that not all morphspaces stretch alike: Only some morphspaces support enhanced discriminability to relevant object dimensions following category learning. Our results have important implications for interpreting neuroimaging studies reporting little or no effect of category learning on object representations in the visual system: Those studies may have been limited by their use of blended morphspaces.

After we learn that objects belong to different categories, do we then perceive those objects differently? And do different kinds of category learning affect visual object perception differently? Someone could learn to categorize mushrooms as edible or poisonous, or as native or invasive species. By some accounts, *how* we categorize objects does not matter: Any experience with mushrooms would facilitate visual perception of mushrooms in exactly the same way (e.g. Op de Beeck & Baker, 2009; Riesenhuber & Poggio, 1999). Other accounts suggest it does matter (Gauthier & Tarr, 2002; Palmeri & Gauthier, 2004).

For example, category learning can selectively enhance the ability to perceive differences along object dimensions, “stretching” dimensions most relevant for categorization (Goldstone, 1994; Gureckis & Goldstone, 2008). This selective stretching can apply *globally* to the entire relevant dimension (Goldstone, 1994; Op de Beeck, Wagemans, & Vogels, 2003) or *locally* to a restricted area around the category boundary (Goldstone, Steyvers, & Larimer, 1996; Notman, Sowden, & Özgen, 2005; Özgen & Davies, 2002; but see Gureckis & Goldstone, 2008 for a different kind of local stretching). Mushrooms not only belong to different categories, but they look more different as a consequence of category learning.

In some cases, the object dimensions exist psychologically before any category learning has occurred. Goldstone (1994), for instance, trained subjects to categorize simple stimuli that varied along the dimensions of size and brightness. While stretching of the relevant size or brightness dimension was observed after category learning, with differences along the category relevant dimension more perceptually discriminable than differences along the irrelevant dimension, it is clear that the psychological dimensions of size and brightness existed long before the experiment. In other cases, the dimensional structure of the objects is far less apparent, and some have suggested that an important component of category learning involves creating

functional features that more clearly distinguish objects from contrasting categories (e.g. Schyns, Goldstone, & Thibaut, 1998). Goldstone and Steyvers (2001), for instance, showed that prior to category learning, a space of morphed faces appeared to have little obvious dimensional organization. After category learning, however, the space of faces appeared to have acquired a set of orthogonal psychological dimensions: one parallel to the learned category boundary and the other perpendicular to the boundary. Therefore, a dimension that becomes stretched through category learning may have existed beforehand, but it may also be created during category learning (Gureckis & Goldstone, 2008).

One inviting interpretation of this body of results is that the visual system has enormous generative power and flexibility, changing and adapting representations of objects to enhance learned category distinctions. However other results do not support this view. Jiang et al. (2007), for instance, found that category learning did little to alter perception of a space of morphed cars. Unlike in Goldstone and colleagues' studies, category learning resulted in equal increases in discriminability along "relevant" morphlines that crossed the category boundary and "irrelevant" morphlines that did not. Object representations are not stretched along category relevant dimensions (Jiang, et al., 2007). In addition, Op de Beeck et al. (2003), only found stretching for simple dimensions that were already psychologically separable before the experiment, and did not find evidence for any creation of new dimensions as a consequence of category learning using more complex objects defined by radial frequency components. These results support a particular theoretical position that how we categorize objects does not matter to how those objects are visually represented (Op de Beeck & Baker, 2009; Riesenhuber & Poggio, 1999).

This debate plays out in neuroimaging and neurophysiological studies. Findings have been mixed regarding the impact of category learning on object representations in visual cortex.

Some studies have found enhanced neural tuning in visual cortex for category-relevant compared to category-irrelevant dimensions while subjects actively categorized objects (De Baene, Ons, Wagemans, & Vogels, 2008; Li, Ostwald, Giese, & Kourtzi, 2007; Sigala & Logothetis, 2002). However, tuning along relevant object dimensions has not been observed outside of active object categorization. While category learning may generally sharpen overall neural tuning for experienced objects, the response patterns of visual neurons do not appear to depend on learned categories, which may be represented in prefrontal cortex and other brain areas instead (Gillebert, Op de Beeck, Panis, & Wagemans, 2008; Jiang, et al., 2007; van der Linden, van Ruenout, & Idefrey, 2010; Knoblich, Freedman, & Riesenhuber, 2002).

So we are faced with mixed behavioral and neural results regarding the apparent flexibility of the visual system in representing objects from learned categories. One line of work suggests stretching along relevant dimensions that already existed or that were created as a consequence of category learning. The other line of work suggests more static object representations that are not tuned to learned categories *per se*. In this article, we propose that different results may have been observed in different studies because qualitatively different kinds of object spaces were used. Our initial focus contrasts a set of studies by Goldstone (e.g., Goldstone, 1994; Gureckis & Goldstone, 2008) with those of Jiang et al. (2007), but our results impact a far broader span of the literature.

Why does category learning affect object perception in some cases (e.g., Goldstone, 1994; Gureckis & Goldstone, 2008) but not others (e.g., Jiang, et al., 2007)? The leftmost panels of Figure 1 (panels a and d) show example object spaces used in those studies. Both depict continuous shape spaces constructed by morphing four parent objects. On the surface, the most obvious difference between the two spaces is the use of faces vs. cars. While this could be a

potentially important difference, because faces are objects of expertise (Tanaka, 2001) and have social relevance, other properties could also be important. For example, Op de Beeck et al. (2003) suggested that category learning enhances perception of relevant dimensions when the space is defined by perceptually separable dimensions, but not when those dimensions are perceptually integral (Garner, 1974). But it is unclear why the faces used by Goldstone et al., who found evidence for dimensional stretching, would be more perceptually separable than the cars used by Jiang et al., who did not. If anything we might expect the opposite, that faces are perceived in a more integral fashion than cars, because faces are known to be processed more holistically than most other objects (Farah, Wilson, Drain, & Tanaka, 1998; Young, Hellawell, & Hay, 1987)ⁱ.

Here we test another explanation for why apparently similar studies obtain qualitatively different results. We demonstrate that the same (non-face) objects can produce dimensional stretching or not depending on how morphspaces are created. In a nutshell, while the two morphspaces illustrated in the leftmost panels of Figure 1 seem to have remarkable similarity, with two axes defined by four parents, they turn out to be qualitatively quite different, as we discuss in the next section. Following a discussion into morphspace creation, our first experiment contrasts the effects of category learning on object perception for these two kinds of morphspaces. Note that while Goldstone et al. used faces and Jiang et al. used cars, our experiments always used cars generated from the same four morphs parents for both morphspaces in both experiments. Experiment 1 shows that category learning causes stretching along relevant dimensions for one kind of morphspace – the “factorial” space used by Goldstone et al. – but not the other – the “blended” space used by Jiang et al. One explanation for this result

is that category learning creates a dimensional structure for the factorial space, but not the blended space. Experiment 2 shows that this is the case.

Morphing is a process by which multiple object images (*parents*) are blended to create a continuum of intermediate object images (*morphs*) and has emerged as an important stimulus generation tool for understanding perceptual categorization. First consider the morphspace shown in Figure 1a, used by Goldstone and Steyvers (2001). As illustrated in Figure 1b, four parents define the space. A morph between Parent A and Parent B define the X Axis of the space; a morph between Parent C and Parent D define the Y Axis of the space. While these axes define the morphspace, they may or may not correspond to the psychological dimensions subjects use to represent the objects within the morphspace, a point we address directly in Experiment 2. The full space is populated with stimuli by morphing images factorially along the X and Y Axes. As a result, we refer to spaces of this type as *factorial* morphspaces. A key property of factorial morphspaces is that they have a consistent structure throughout the space. Any two stimuli in the space that differ along the X Axis but are equated along the Y Axis generally differ in the degree to which they are physically similar to Parent A vs. Parent B, regardless of where they are located along the Y Axis. Similarly, any two stimuli that differ along the Y Axis but are equated along the X Axis are generally equal in their physical similarity to Parent A vs. Parent B, regardless of where they are located along the X Axis.

Now consider the morphspace shown in Figure 1d, used by Jiang et al. (2007; see also Freedman et al., 2003; Gillebert et al., 2008). On the surface, this morphspace looks structurally identical to the dimensional morphspace discussed above. But the two-dimensional layout in the figure masks a fundamental structural difference. In this case, a unique parent occupies each corner of the space (Figures 1d and 1e). These parents occupy the most extreme points in a three

dimensional tetrahedron, not a two-dimensional plane. The morphlines depicted in Figure 1d and 1e are merely the edges of that tetrahedron projected onto a plane. The space used by Jiang et al. (2007) is populated with objects by sampling from all possible blends of the four parents. In this way, a given object's position within the space represents the relative morphing contribution from each of the four parents to the object. Objects close to parent A are blends of all four parents with a dominant contribution from parent A; objects lying on one of the four outer walls of the pyramid are blends between three of the parents with no contribution from the fourth parent on the corner opposite that wall; objects lying on one of the edges of the pyramid are blends between two parents with no contribution from the other two. We will refer to spaces of this type as *blended* morphspaces. A key property of blended spaces is not that stimuli necessarily occupy all regions of the tetrahedron, but that the corners of the blended space correspond to the morphparents at the vertices of the tetrahedron. So despite the striking similarity in how they are typically depicted in Figures 1a and 1d, the blended space defines a three-dimensional tetrahedron while the factorial space defines a two-dimensional plane. In addition, the blended space and factorial spaces differ significantly with respect to how the stimuli within the spaces relate to the learned category boundary, as described next.

Consider how categories are defined in the two morphspaces. We have depicted a “vertical” category boundary for both spaces in Figure 1 but the nature of those boundaries is actually quite different across the two. For the factorial space, the difference between Category 1 and Category 2 lies in the relative contribution of Parent A or Parent B along morphline Axis X, regardless of the relative contribution of Parent C or Parent D along morphline Axis Y. This opens up the possibility that category learning will encourage attention to the degree to which a given object resembles Parent A vs. Parent B, while the resemblance to Parent C vs. Parent D

will be ignored. This could selectively enhance visual representations of object variability along the X Axis, even though this variability could be complex and difficult to verbalize.

This contrasts with the blended space where the categorical distinction is more complex. The category boundary is defined by the half-way point along the morphlines between Parents A and C, Parents A and D, Parent B and C, and Parents B and D, and is defined continuously within the tetrahedral space by the plane depicted in Figure 1f. Whatever learned perceptual representations help discriminate stimuli along the A to C morphline will be completely different from those that help discriminate stimuli along the B to D morphline. In contrast to the factorial space, in a blended space there are multiple distinctions between stimuli in the two categories that reflect variations along multiple directions. Perhaps because so many distinctions are relevant to categorization, little or no difference in perceptual discrimination along purportedly relevant versus irrelevant directions in the blended space is observed after category learning.

To our knowledge, this difference between factorial and blended spaces has never been noted in the literature, perhaps because the two spaces have been graphically depicted so similarly in published papers (e.g., leftmost panels of Figure 1). But does this difference matter behaviorally? If it does, this has potentially important implications in interpreting results on the neural substrates of category learning, given the papers that have used blended spaces and concluded that category learning has little systematic influence on object representations in visual cortex (e.g. Freedman, Riesenhuber, Poggio, & Miller, 2003; Gillebert, et al., 2008; Jiang, et al., 2007).

In the first experiment, we directly contrast changes in perceptual discrimination after category learning for stimuli constructed from a factorial morphspace versus a blended morphspace. We constructed subspaces of stimuli with a similar two-dimensional structure and a

similar category boundary, using the same parent objects, but in one case the stimuli were defined within a factorial morphspace while in the other case the stimuli were defined within a blended morphspace (See Figure 2).

EXPERIMENT 1

Methods

Stimuli

Pilot studies using multidimensional scaling were used to select 4 car images to be parents with approximately equal similarity to each other from among 30 total images (selected from a commercial collection of three-dimensional computer models available online at www.doschdesign.com/products/3d/Lo-Poly_Cars_V1-2.html). Once selected, these four parents were further processed using Adobe Photoshop to remove unnecessary details that would have interfered with smooth morphing. The parents are shown in Figure 2c and 2f as parents A, B, C, and D.

The factorial and blended morphspaces were created from the four parents using the approach described earlier. We first note that the factorial space is actually a two-dimensional plane within the three-dimensional tetrahedron defining the full blended space (Figure 1c, 2a). To compare category learning and its impact on perceptual discrimination in the two different morphspaces, we needed to equate the two spaces as much as possible while retaining the key qualitative properties of each morphing method. The twisted two-dimensional manifold within the full blended space possesses key properties of the full blended space. First, a unique parent resides at each corner of the space. Second, a morphline between two parents defines each edge of the space; these morphlines are a subset of those tested by Jiang et al. (2007). Figure 2d shows

the blended space used in Experiment 1 – the space is compressed slightly so that adjacent pairs within the space are as discriminable as the pairs in the factorial space. From now on, *blended space* will refer to this twisted surface within the full tetrahedron. Figure 2b and 2e show “flattened” schematics of the factorial and blended spaces to illustrate the analogous arrangement of the exemplars within the two spaces.

Separate sets of stimuli were created for category learning and perceptual discrimination phases of the experiment, so that the effects of category learning on perception would be tested on transfer objects interspersed within the training space. We used an 8x8 object set for Category learning (shown by filled circles in Figure 2b and 2e) while a non-overlapping 4x4 object set was used in the perceptual discrimination task (shown as open squares in Figure 2b and 2e).

Importantly, if we used the original parents at the corners of the blended space, the average perceptual discriminability would be greater than for the corresponding factorial space (independently of any category learning). Therefore, extensive pilot testing was used to select new corner locations for the blended space to better match perceptual discriminability prior to category learning between the two spaces (Figure 2d).

The exact contribution of each parent to each morphed object in the blended and factorial spaces is available from the authors or online at catlab.psy.vanderbilt.eduⁱⁱ. Objects were produced using `gtkmorph` (xmorph.sourceforge.net).

Procedure

The procedure implemented a 2x2x2 experimental design with factors of Space (factorial vs. blended), Discrimination Epoch (pre-training vs. post- training), and Pair Relevance (relevant vs. irrelevant). Each subject was randomly assigned to perform categorization and discrimination

tasks on either the factorial space or the blended space. In the first experimental session, subjects performed a same-different perceptual discrimination pre-test on pairs of the 4x4 object set, followed immediately by category learning with the 8x8 object set. In the second session, subjects underwent additional category learning with the 8x8 object set followed immediately by a same-different perceptual discrimination post-test on pairs of the 4x4 object set. The sessions were conducted on two consecutive days and each session lasted approximately two hours.

Same-different perceptual discrimination task. In the perceptual discrimination task, subjects judged whether sequentially presented objects, pairs of identical or adjacent cars in the 4x4 grid, were the same or different (even if only slightly). Each trial proceeded as follows: 700ms fixation, 1500ms car image 1, 300ms black and white noise mask, 300ms blank screen, car image 2, which remained on screen until the subject responded “same” or “different” or 5 seconds had elapsed. Inter-trial interval was 1 second. Each block consisted of 24 different trials and 8 same trials in a random order. All 16 possible “same” trials were presented every 2 blocks. There were a total of 12 blocks (384 total discrimination trials per session). The “different” trials consisted of adjacent pairs, cars that differed by a single horizontal or vertical position in the space. There were a total of 24 possible adjacent pairs, 12 of which differed in the vertical direction and 12 of which differed in the horizontal direction. Depending on the orientation of the categorization boundary for a given subject, a pair was considered *relevant* if the difference was perpendicular to the direction of the category boundary or *irrelevant* if the difference was parallel to the category boundary.

Category learning task. In the category learning task, subjects learned to categorize cars in the 8x8 space carved in half according to either a “vertical” or “horizontal” category boundary (counterbalanced across subjects in both morphspace groups). If a given subject learned to

categorize according to a vertical boundary, then horizontal pairs were deemed relevant in the discrimination task; if a horizontal boundary was learned, vertical pairs were deemed relevant. Subjects were instructed to learn to categorize two novel brands of cars made by an imaginary manufacturer as either “Cags” or “Mons”. Each trial proceeded as follows: fixation 500ms, car image 1500ms, feedback 700ms, intertrial interval 1000ms. Feedback consisted of both the correct category name and whether the subject’s response was correct or incorrect, 300ms after the response or 300ms after stimulus offset, whichever was later. All 64 cars in the 8x8 space were presented in each block and there were 12 blocks on each training day for a total of 768 category learning trials per day.

Subjects

Subjects were 24 Vanderbilt students and members of the community, 13 of whom were female, who were reimbursed \$24 (average 22.4 years old, average 15.4 years of education). The study was approved by the local IRB.

Results

Category learning. Subjects were slightly more accurate in categorizing the factorial space than in categorizing the blended space (Factorial: 84.4%, Blended: 75.9%, $F(1,22) = 6.65$, $MSe = .007$, $p = .017$, $\eta^2 = .232$), but the groups did not differ in reaction time (reaction time: $F(1,22) < 1$).

Same-different perceptual discrimination. Perceptual discriminability (Figure 3) improved more for the relevant than irrelevant pairs in the factorial space but not for pairs in the blended space (Relevance x Discrimination Epoch interaction: $F(1,22) = 5.90$, $MSe = .029$, $p =$

.024, $\eta^2 = .212$; Relevance x Discrimination Epoch x Space interaction: $F(1,22) = 7.58$, $MSe = .029$, $p = .012$, $\eta^2 = .256$). Planned comparisons of the increase in discriminability (d') after category learning for relevant compared to irrelevant pairs confirmed this impression. In the factorial space, the increase in d' after category learning was greater for relevant pairs compared to irrelevant pairs ($t(11) = 2.90$, $p = .014$, Cohen's $d = .860$), while, in the blended space, d' increased equally for both relevant and irrelevant pairs ($t(11) = -0.36$, $p = .7$, Cohen's $d = -.104$)ⁱⁱⁱ. Overall, subjects were better at discriminating between adjacent object pairs after categorization than before categorization (main effect of Discrimination Epoch: $F(1,22) = 4.71$, $MSe = .129$, $p = .041$, $\eta^2 = .176$). The initial discriminability of pairs in the factorial and blended spaces were equal, as revealed by an ANOVA on the pre-categorization discrimination epoch alone with factors of Space and Relevance; the main effect of Space was not significant ($F(1,22) = .067$, $MSe = .150$, $p = .798$, $\eta^2 = .003$) and the effects of Relevance and of Space by Relevance were also not significant (Relevance: $F(1,22) = 1.30$, $MSe = .052$, $p = .27$, $\eta^2 = .056$; Space by Relevance: $F(1,22) = 1.69$, $MSe = .052$, $p = .21$, $\eta^2 = .071$).

We found increases in discriminability for relevant pairs in the factorial space but not the blended space. We can also ask whether this modulation was confined to the local region near the category boundary or applied globally to the entire relevant direction in the space. Figure 4 shows d' for relevant and irrelevant pairs broken down by whether or not they crossed the middle of the object space. Relevant pairs crossed the category boundary, while the non-relevant pairs crossed the middle of the space in the irrelevant direction. For both spaces, pairs that crossed the middle of the space increased in discriminability more than those that did not, regardless of relevance. In an ANOVA with factors of Space, Relevance, Discrimination Epoch, and Pair-Position (whether a given pair spanned the center of the space), we found a main effect of Pair-

Position ($F(1,22) = 7.75$, $MSe = .039$, $p = .011$, $\eta^2 = .261$) and a Pair-Position by Discrimination Epoch interaction ($F(1,22) = 6.52$, $MSe = .037$, $p < .018$, $\eta^2 = .229$). No interactions between Pair-Position and Space or Pair-Position and Relevance were significant ($F_s < 1$). Thus, discriminability effects specific to the relevant dimension applied globally to the relevant dimension and were not restricted to pairs that crossed the category boundary. All interactions and main effects not involving Pair-Position replicated those in the previous analysis and are not reported here for sake of brevity.

Discussion

Goldstone and colleagues observed that category learning causes increases in perceptual discriminability along category-relevant object dimensions (Goldstone, 1994; Goldstone & Steyvers, 2001; Gureckis & Goldstone, 2008). Recent behavioral and neural studies suggest otherwise (Freedman, et al., 2003; Gillebert, et al., 2008; Jiang, et al., 2007; van der Linden, et al., 2010). We provide one explanation for these conflicting results in terms of how the morphspaces of stimuli used in these studies were constructed.

We used the same parent objects to create factorial (Gureckis & Goldstone, 2008) and blended (Jiang et al., 2007) morphspaces and investigated the effect of category learning on perceptual discriminability within each space. For the factorial morphspace, there was enhanced discriminability along the category-relevant dimension. This was not observed for the blended morphspace. Morphspace structure must be taken into account when interpreting effects of category learning on the representation of objects. Not all morphspaces stretch alike.

The results of Experiment 1 suggest that there is something about the structure of factorial spaces, but not blended spaces, that allows selective increases in discriminability along relevant dimensions. But what? We address that question in the next experiment.

EXPERIMENT 2

In Experiment 2, we investigated possible explanations for how and why the factorial morphspace allows stretching along the relevant dimension following category learning while the blended space does not. Specifically, we assessed the extent to which objects in each space are represented in terms of psychological dimensions. In a space that has a dimensional structure, enhanced perceptual discriminability following category learning can be explained as stretching along the relevant psychological dimension. The fact that the factorial space is created by factorially combining morphs across two independent morphline axes and the observation that stretching along the relevant dimension was observed suggests a dimensional structure for the factorial space. Furthermore, the observation that category learning did not lead to stretching in the relevant direction for the blended space suggests that the psychological representations of objects in the space do not correspond directly to the axes that define the space. Here we test these intuitions explicitly.

Following Goldstone and Steyvers (2001), our approach relies on the assumption that if a space has a psychological dimensional structure, then it should be easier to learn categories that vary along a single dimension than categories that vary along two dimensions (e.g. Gottwald & Garner, 1975). This is because single-dimension categories benefit from stretching along the relevant dimension (e.g. Nosofsky, 1986; Nosofsky, Gluck, Palmeri, McKinley, & Gauthier, 1994; Kruschke, 1992, 1993). By contrast, if the dimensions are completely arbitrary, there

should be no difference in how easy or hard it is to learn categories that vary along one dimension versus two dimensions, because the “dimensions” are merely a fiction of how the space of objects is created or displayed graphically (see Op de Beeck et al., 2003 for a slightly different perspective).

Consider the two-dimensional space of objects illustrated in panel a of Figure 5. The X and Y Axes are defined by the way the morphspace is created. If these axes correspond to psychological dimensions that can be selectively stretched when relevant, as we hypothesize for the factorial morphspace, then it should be relatively easy to learn the categories illustrated in panel b: The relevant X Axis corresponds to a psychological dimension that can be stretched. By contrast, it should be relatively hard to learn the categories illustrated in panel c because both dimensions are relevant: Stretching is not possible in the diagonal direction.

By contrast, if the morph axes are completely arbitrary, with no correspondence to psychological dimensions, as we hypothesize for the blended morphspace, there should be no difference in learning between the categories illustrated in panels b and c. The X and Y Axes have no special psychological status in that the axes can be rotated by any degree, with every rotation an equally valid description of the space of morphed objects. With no psychological dimensions, nothing can be stretched.

Hopefully, it should be clear from this discussion that the seemingly complicated tessellation of the object space into nested octagons in Figure 5 has a purpose: This octagonal arrangement allows us to rotate the category boundary by 45 or 90 degrees yet preserve the relative relationships of objects within and across the two categories (Kruschke, 1993; see also Shepard, 1964; Nosofsky, 1985). Contrast this with the square configuration used in Experiment 1, Figure 2, which cannot be rotated by 45 degrees.

Experiment 2 included two phases of category learning, to test two hypotheses about when, if ever, a dimensional structure might emerge within a morphspace.

One explanation for the differences observed in Experiment 1 is that the factorial morphspace has a dimensional structure that exists prior to any category learning, with morph axes corresponding to psychological dimensions, while the blended space does not. According to this hypothesis, category learning in the factorial morphspace would have simply caused stretching along a preexisting relevant dimension, thereby making relevant distinctions more perceptually discriminable (Goldstone, 1994; Op de Beeck et al., 2003). In Phase One of Experiment 2, the presence of a preexisting dimensional structure in either space should be evidenced by easier learning of the orthogonal category boundary (panel b) than the diagonal category boundary (panel c).

Another explanation for the differences observed Experiment 1 is that the factorial morphspace allows a dimensional structure to be created during category learning, while the blended space does not. By this account, category learning involves creating the right dimensional structure for the object categories being learned (Schyns, et al., 1998). This would suggest that in Experiment 1, for a dimension to be stretched, the dimension must first be created. For example, Goldstone and colleagues (Gureckis & Goldstone, 2008; Goldstone & Steyvers, 2001; see also Austerweil & Griffiths, 2010) found that after learning to categorize a morphspace of faces, the faces were embedded in a dimensional structure consistent with the learned categories, a dimensional structure that did not exist prior to category learning. In Phase Two of Experiment 2, we use a procedure that can detect evidence of a learned dimensional structure: we outline this procedure next.

Imagine that after learning the category structure illustrated in panel d of Figure 5, the space of objects is defined by the Relevant and Irrelevant Dimensions illustrated. Now, for the panel d structure these dimensions correspond to X and Y Axes that defined the original morphspace. Now consider the category structure illustrated in panel e. Again, imagine that after learning, the space of objects is defined by the Relevant and Irrelevant Dimensions illustrated. Of course, now the dimensions do not correspond to the original X and Y Axes, but the Relevant and Irrelevant Dimensions are both psychologically real.

For both cases, now consider what happens when a new set of categories must be learned using objects within the same space. The new category boundary could be defined by a 90-degree rotation (panels g and i) or by a 45-degree rotation (panels f and h) of the original category boundary. Which rotation will be easier to learn? If the Relevant and Irrelevant dimensions are psychologically real as a result of Phase One of category learning, then the 90-degree rotation category should be easier to learn than the 45-degree rotation category. That is because the 90-degree rotation category relies on the original dimensions, only switching which one is relevant. By contrast, the 45-degree rotation creates a situation where both of the original dimensions are relevant. Either both dimensions must be used, which is less efficient than having a single relevant dimension, or a new dimensional structure must be learned, which is less efficient than simply switching the relevant and irrelevant dimensions. Either way, if Phase One category learning creates a dimensional structure, then the 90-degree rotation category should be easier to learn than the 45-degree rotation category.

To summarize, subjects in Experiment 2 learned two categorizations of the same morphspace of objects. Half of the subjects learned objects from the factorial space, and half from the blended space. In Phase One, we tested whether category learning is more efficient for

categories that vary along one of the axes defined in the morphspace (“orthogonal”), than for categories that vary along both axes (“diagonal”). Phase One lets us test whether or not a morphspace had a preexisting dimensional structure that maps onto the morph axes. In Phase Two, we transferred subjects to a new category whose boundary varies by either 90 degrees or 45-degrees from the original boundary. If Phase One led subjects to create a dimensional structure in the morphspace, we expect faster learning for the 90-degree rotation category than the 45-degree rotation category.

Methods

Stimuli

We used the same four parent morphs and morphing techniques used in Experiment 1. The only difference was the x,y locations of morphs sampled from each space. While Experiment 1 used a square grid, Experiment 2 instead used an octagonal tessellation to allow for a fair comparison of 45-degree and 90-degree rotations (see Figure 6). The blended space was scaled on the basis of pilot testing to equate the difficulty of category learning as much as possible with the factorial space. The exact coordinates of the stimuli illustrated in Figure 6 are available from the authors or online as supplementary material at catlab.psy.vanderbilt.edu.

Procedure

Subjects were randomly assigned to categorize the factorial or blended morphspace. Each subject learned to categorize the assigned space according to two different category boundaries in two phases of category learning.

In Phase One, subjects were randomly assigned to learn categories defined by either a boundary that was orthogonal to one of the axes of the morphspace (panel b of Figure 5) or by a boundary that was diagonal to both of the axes of the morphspace (panel c of Figure 5). For the orthogonal condition, whether the boundary was orthogonal to the X Axis or Y Axis was counterbalanced across subjects; analogously, for the diagonal condition, whether the boundary had a positive or negative slope was counterbalanced across subjects. Relating this to Experiment 1, for both the blended and factorial spaces, in the orthogonal category learning condition the category boundary was orthogonal to the relevant dimension, while in the diagonal condition the category boundary was not. In Experiment 2, comparing the efficiency of category learning in the orthogonal versus diagonal category learning conditions allows us to assess whether the two spaces differ in their a priori dimensional structure. The category learning instructions for Phase One were identical to those used in Experiment 1.

In Phase Two, subjects were instructed that the cars had changed manufacturers and that they would then have to learn to categorize the cars in a new way. Subjects learned to categorize the same cars according to a boundary that was rotated either 90 degrees or 45 degrees (randomly assigned) from the original orientation used in Phase One. The direction of the rotation was counterbalanced across subjects. Examples of 90 degree and 45 degree rotations are shown in Figure 5.

Phase One took place over two consecutive days, with 792 category learning trials, distributed over 11 training blocks, on the first day and 792 additional trials on the second day. Phase Two was conducted on the second day immediately following the end of Phase One and consisted of 792 trials during which subjects were trained on the new boundary. The sequence

and timing of the stimuli and feedback was identical to the category learning phase of Experiment 1.

Subjects

A total of 64 members of the Vanderbilt community, 40 female, participated in Experiment 2. An additional 8 subjects were dropped due to technical problems or failure to complete the entire experiment. 32 subjects were trained to categorize each space (factorial or blended), 16 of which were first trained to categorize their space according to a diagonal boundary and 16 of which were trained to categorize their space according to an orthogonal boundary. For each of these groups of 16 in Phase Two, 8 subjects were trained on a new boundary rotated 45 degrees relative to the original boundary while the remaining 8 subjects were trained on a new boundary rotated 90 degrees relative to the original boundary.

Results

Phase One: Orthogonal versus Diagonal Category Learning. Figure 7 displays the category learning curves for Phase One. To investigate the effect of diagonal vs. orthogonal category boundaries on overall accuracy and reaction time, we first conducted ANOVAs with factors of Training Block (22 total blocks), Boundary Orientation (diagonal vs. orthogonal), and Space (Factorial vs. Blended) with violations of sphericity corrected using the Huynh-Feldt method. Significant main effects of Training Block showed that subjects became more accurate and responded more quickly as training progressed (accuracy: $F(21,1260) = 34.1$, $MSe = .009$, $p < .0005$, $\eta^2 = .362$; reaction time: $F(21,1260) = 7.5$, $MSe = .093$, $p < .0005$, $\eta^2 = .111$). In the remaining analyses, the effects on reaction time were either not significant or mirrored the effect

of accuracy, so they are not considered further. There was no overall difference in accuracy between orthogonal and diagonal category boundaries (main effect of Boundary Orientation: $F(1,60) = 1.56, MSe = .174, p = .217, \eta^2 = .025$), nor did the main effect of Boundary Orientation differ between the blended and factorial spaces (Orientation x Space interaction: $F(1,60) = .019, MSe = .174, p = .892, \eta^2 < .0005$). There was also no overall difference in learning rate between orthogonal and diagonal boundaries (Orientation x Block interaction: $F(21,1260) = .571, MSe = .009, p = .810, \varepsilon = .399, \eta^2 = .009$) and the difference in learning rate between boundary orientations did not interact with Space (Orientation x Space x Block interaction: $F(21,1260) = 1.44, MSe = .009, p = .18, \varepsilon = .399, \eta^2 = .023$)^{iv}. While there was no overall difference in accuracy between the two spaces (main effect of Space: $F(1,60) < 1$), Space did interact with Block, likely reflecting slightly higher accuracy for the blended space in the first two training blocks (Space x Block interaction $F(21,1260) = 1.97, MSe = .009, p = .045, \varepsilon = .399, \eta^2 = .032$).

In summary, we found little evidence for an advantage of learning the orthogonal boundary over the diagonal boundary for either space. Neither space appears to have a strong dimensional structure prior to category learning, ruling out differences in preexisting dimensional structure between spaces as an account for the differences observed in Experiment 1. We now turn to an analysis of Phase Two to see if a dimensional structure could have emerged in either space as a result of category learning.

Phase Two: 90-degree versus 45-degree Category Learning. To test whether category learning caused dimensions to emerge, we compared accuracy during Phase Two of category learning when boundaries were rotated either 90 degrees or 45 degrees from the original boundary. This comparison is complicated by a difference between the 90- and 45-degree

rotation conditions, namely that more objects change categories in the 90-degree than the 45-degree rotation condition, as shown in Figure 8. To compensate for this difference, we analyzed only the subset of exemplars in each space that changed categories in the two conditions and that were matched in distance from the new category boundary. These critical exemplars for the example conditions shown in Figure 5 are shown as darker dots.

Figure 9 shows the learning curves for the critical exemplars, for category boundaries rotated by 90 or 45 degrees. In the factorial space, learning appears faster after the 90- than 45-degree rotation, while there is little apparent difference in the blended space. We first analyzed accuracy in both spaces with an omnibus ANOVA with factors of Training Block (11 total blocks), Boundary Rotation (45 degrees vs. 90 degrees), and Space (Factorial vs. Blended). A significant main effect of Block showed that subjects learned over time ($F(10,600) = 7.40$, $MSe = .022$, $p < .0005$, $\epsilon = .775$, $\eta^2 = .110$). The only other significant effect was a three-way interaction between Block, Space, and Boundary Rotation, suggesting that the effect of boundary rotation on learning rate differed between the two spaces ($F(10,600) = 2.50$, $MSe = .022$, $p = .013$, $\epsilon = .775$, $\eta^2 = .040$).

To follow up on this significant interaction, we performed separate ANOVAs within each space with factors of Block and Boundary Rotation. In the factorial space, learning rate was significantly slower for the 45-degree rotation than the 90-degree rotation (Figure 9a) (Rotation x Block interaction: $F(10,300) = 2.34$, $MSe = .028$, $p = .026$, $\epsilon = .68$, $\eta^2 = .072$). In contrast, the learning rate did not vary significantly with boundary rotation in the blended space (Figure 9b; Rotation x Block interaction: $F(10,300) = 1.38$, $MSe = .017$, $p = .197$, $\epsilon = .92$, $\eta^2 = .044$). The main effect of Rotation was not significant in either space ($F(1,30) < 1$) and the main effect of

Block was significant in both spaces (Factorial: $F(10,300) = 3.26$, $MSe = .028$, $p = .003$, $\varepsilon = .69$, $\eta^2 = .098$; Blended: $F(10,300) = 5.15$, $MSe = .017$, $p < .0005$, $\varepsilon = .92$, $\eta^2 = .15$).

DISCUSSION

Experiment 2 asked whether the factorial and blended morphspaces contained a psychological dimensional structure, either prior to any category learning, or as a result of category learning. Specifically, if the factorial morphspace contains a dimensional structure, with morph axes corresponding to psychological dimension, but the blended morphspace does not, that would help explain why stretching along the relevant dimension was observed in Experiment 1 for the factorial space but not the blended space. Increases in perceptual discriminability along category relevant dimensions may only be possible when a dimensional structure exists.

In Phase One of Experiment 2, neither space showed a significant advantage for learning the orthogonal categories, defined along only one of the morph axes, compared to the diagonal categories, defined along both morph axes. This suggests that in both spaces, the particular axes that defined the morph spaces were not more psychologically real than a pair of diagonal dimensions.

In Phase Two of Experiment 2, we observed an advantage when subjects learned a second category when the new category boundary was defined by a 90-degree rotation compared to a 45-degree rotation for the factorial morphspace but not for the blended morphspace. Following Goldstone and colleagues (Goldstone & Steyvers, 2001; Gureckis & Goldstone, 2008) this implies that Phase One category learning created a dimensional structure within the factorial morphspace, but not the blended morphspace. With a dimensional structure in the factorial

morphspace, stretching along a relevant dimension is possible, leading to the enhanced perceptual discriminability following category learning observed in Experiment 1.

GENERAL DISCUSSION

There have been conflicting results regarding the effect of category learning on visual representations. Some studies have found increased discriminability and sharper neural tuning for features along relevant dimensions while other studies have found no difference. For instance, after learning to categorize a continuous morphspace of novel cars, Jiang et al. (2007) found no difference in perceptual discriminability along a category-relevant versus category-irrelevant dimension. This contrasts with other behavioral work showing enhanced perceptual discriminability along relevant dimensions after learning to categorize simple shapes varying in size and brightness (Goldstone, 1994), colors varying in brightness and saturation (Goldstone & Styvers, 2001), and unfamiliar faces varying across a continuous morphspace (Gureckis & Goldstone, 2008). One explanation for the conflicting behavioral results is that different studies used different techniques for creating morph spaces.

We measured the effect of category learning on perceptual discrimination of pairs of car images that differed along either the relevant or irrelevant direction sampled from factorial morphspaces (e.g., Goldstone & Steyvers, 2001) or from blended morphspaces (e.g., Jiang et al., 2007). In the factorial space, but not the blended space, category learning resulted in increased discriminability along the relevant dimension. These findings highlight a potentially critical factor in the methodology of experiments testing for effects of category learning on object representations, one that has not been acknowledged up to now. These results may help explain why some studies have obtained selective increases in discriminability along relevant dimensions

following category learning (Goldstone et al., 2001; Gureckis & Goldstone, 2008): They used factorial spaces. Other studies have not observed increases in discriminability specific to the relevant dimension (Gillebert et al., 2008; Jiang et al., 2007): They used blended spaces.

Our findings with the factorial space are broadly consistent with those of Gureckis and Goldstone in that we observed an increase in discriminability specific to the relevant dimension.^v Our findings are also consistent with Goldstone and Steyvers (2001) in that category learning caused the development of a dimensional structure. But the development of a dimensional structure was not omnipresent (Op de Beeck et al., 2003) since we observed it for the factorial but not the blended morphspace. This appears to support a middle ground between a view that dimensional stretching can only be observed with simple, separable object dimensions and a view that the visual system can impose a dimensional structure on any complex shape space (Goldstone & Steyvers, 2001; Hockema, Blair, & Goldstone, 2005)^{vi}.

Overall, it is clear that learning to categorize objects generated by blended vs. factorial morphing methods results in different visual representations, even after only two one-hour training sessions. It is possible that following more extensive training, other qualitative differences could emerge besides the dimensional relevance effect we observed. In particular, categorical perception – a sharpening of representations near the boundary only along the relevant dimension – might be observed in behavior (and brain activity) after sufficient amounts of training with a factorial morphspace. Prior work suggests that this would not be obtained in a blended morphspace (Freedman et al., 2003; Gillebert et al., 2008; Jiang et al., 2007).

Our results not only put previous behavioral results under a new light, but also impact our understanding of how objects are represented in the brain and how category learning affects those object representations. For example, Jiang et al. (2007) trained subjects to categorize

objects from a space of morphed cars and measured fMRI adaptation to pairs of cars before and after training. Greater release from adaptation after training would indicate increased separability between neural populations representing similar cars. However, in object-sensitive areas of visual cortex, Jiang et al. found as much adaptation for pairs of cars in different categories as for pairs of cars within the same category, suggesting no enhancement in neural representations around the category boundary. Using analogous methods, Gillebert et al. (2008) and van der Linden et al. (2010) also found no evidence for systematic changes in neural discriminability in object-sensitive regions of visual cortex as a result of category learning. Overall, such findings are consistent with a view that visual areas represent objects but categories are represented in other brain areas (e.g. Freedman & Miller, 2008; Seger & Miller, 2010; Serre, Oliva, & Poggio, 2007). While object representations in visual cortex could develop overall sharper tuning with experience, by this view, these experience-dependent changes are insensitive to the type of experience, such as whether particular object dimensions are relevant or irrelevant to previously learned categories.

Importantly, studies reporting no evidence for category-related effects on object representations in the visual system also did not report commensurate behavioral evidence for changes in perceptual discriminability. Either no behavioral measures were made or those behavioral measures returned null results. It seems a mistake to interpret those null results as a universal characterization of how category learning affects object representations. Indeed, we argue that no behavioral effects were observed because those studies used blended morphspaces.

When objects have some kind of dimensional structure, category-related changes in perceptual discriminability are observed. Category-related changes in neural sensitivity in visual cortex are too. For example, De Baene et al. (2008) trained monkeys to categorize a stimulus

space defined by curvature and aspect ratio and found clear increases in neural sensitivity to the relevant dimension in IT (see also Sigala & Logothetis, 2002). The same study also found increases in similarity in neural responses to objects in the same category compared to objects in different categories. However, in these experiments, monkeys were only tested during a categorization task, leaving the question open as to whether selective attention during category learning altered neural sensitivity dynamically rather than in a stable, task-independent fashion (Gauthier & Palmeri, 2002).

We recently trained subjects to categorize objects from the same factorial space used in Experiment 1 and then measured fMRI adaptation to sequentially presented pairs of objects differing along a relevant or irrelevant dimension during a match-to-location task that did not require any categorization. Our design was similar to Jiang et al. (2007) but used a factorial morphspace rather than a blended morphspace. Consistent with the behavioral results observed in the current article, we found that pairs of objects differing along the relevant dimension showed less fMRI adaptation than irrelevant pairs in object-sensitive areas of ventral temporal cortex (Folstein, Palmeri, & Gauthier, under review). Because we used a match-to-location task, we can conclude that the effect was not dependent on active categorization. Category learning systematically affects object representations in visual cortex in more ways than simply through a global sharpening of object representations.

Clearly, changes in behavioral and neural sensitivity to objects must be examined in the context of the specific stimulus properties used to test for those changes. Op de Beeck et al. (2003) suggested that increased sensitivity along a dimension relevant for categorization will only be found for relatively separable dimensions, like the curvature and aspect ratio used by De Baene et al. (2008), the distinct object parts used by Sigala and Logothetis (2002), or the shape

and motion used by Li et al. (2007). Unlike these simple objects, recent studies using complex multidimensional morphed objects (Freedman et al., 2003; Jiang et al., 2007; Op de Beeck et al., 2003) have concluded that effects of category learning on visual object representations do not depend on category relevance. Our results suggest that any effect of category learning on perception of complex objects defined within morphspaces is highly dependent on the structure of those morphspaces. Category learning can selectively enhance object representations along relevant dimensions in a continuous shape space, at least when the shape space is defined factorially.

NOTES

ⁱ Another way that studies that find or do not find dimensional stretching differ is in the tasks performed by the subjects in the context of category learning. Studies using blended spaces have sometimes used matching tasks to train their subjects (Jiang et al., 2007, Freedman et al., 2003) rather than tasks designed to associate categories with buttons or labels (Goldstone 1994, Goldstone and Steyvers 2001). Differences in the degree to which different category learning tasks cause dimensional stretching is a potentially interesting topic for future studies.

ⁱⁱ http://catlab.psy.vanderbilt.edu/wp-content/uploads/JEPLMCMAN4_website_Final.pdf

ⁱⁱⁱ Categorization accuracy was somewhat lower in the blended space than the factorial space, raising the possibility that the difference we observed between groups could be related to something like participant motivation. To investigate this, we removed the three participants in the blended group with the lowest categorization accuracy and the three participants in the factorial group with the highest categorization accuracy, leaving 9 participants in each group. When doing this, the difference in accuracy between the two groups no longer approached significance (factorial $M = 82\%$, blended $M = 80\%$; $F(1,16) = .545$, $MSe = .003$, $p = .47$, $\eta^2 = .033$). Even when the groups were equated for categorization accuracy the critical interactions for the discrimination phase remained significant (Diagnosticity x Epoch: $F(1,16) = 8.29$, $MSe = .024$, $p < .05$; Diagnosticity x Epoch x Space: $F(1,16) = 8.73$, $MSe = .024$, $p < .01$, $\eta^2 = .353$). Thus the difference between the spaces in factorial modulation results from the properties of the spaces and not random variability between participants.

^{iv} To increase the sensitivity of our analysis, we also conducted separate exploratory two-way Orientation x Block ANOVAs on each space to search for evidence of differences between diagonal and orthogonal boundaries. In the factorial space, the main effect of block was again significant ($F(21,630) > 20$), but there was no difference in accuracy between orthogonal and diagonal boundaries ($F(1,30) = .56$, $MSe = .191$, $p = .46$, $\eta^2 = .018$), nor was there a significant difference in learning rate (Orientation x Block interaction: $F(21,630) = .776$, $MSe = .013$, $p = .614$, $\varepsilon = .35$, $\eta^2 = .025$). To further maximize sensitivity to effects involving Block, we subsequently performed a trend analysis with Block as an ordered factor, breaking down the effect of block into orthogonal trends with one degree of freedom each. Neither the linear nor the quadratic trends were significant for the Block by Orientation interaction (linear: $F(1,30) = 1.40$, $MSe = .022$, $p = .25$, $\eta^2 = .045$; quadratic: $F(1,30) < 1$).

In the blended space, there was also no difference in accuracy between the two boundary orientations (main effect of Boundary Orientation: $F(1,30) = 1.063$, $MSe = .157$, $p = .31$, $\eta^2 = .034$), nor was there any significant difference between the learning rates in these conditions (Orientation by Block interaction: $F(21,630) = 1.37$, $MSe = .006$, $p = 1.96$, $\eta^2 = .044$). Finally, the effect of Boundary Orientation did not interact with the linear or quadratic trends of Block (linear: $F(1,30) = 1.30$, $MSe = .012$, $p = .26$, $\eta^2 = .041$; quadratic: $F(1,30) = 1.03$, $MSe = .006$, $p = .32$, $\eta^2 = .033$).

Visual inspection of Figure 8 suggests that, for the blended space, the orthogonal boundary appears to have a slight accuracy advantage late in learning, after block 8. By contrast, for the factorial space, the orthogonal boundary appears to have a slight accuracy advantage early in learning, from blocks 1 through 5. We therefore performed a completely post hoc analysis of these epochs in isolation for the two spaces. In each space two ANOVAs were

performed with factors of Block and Boundary Orientation. In one ANOVA, Block had five levels, corresponding to blocks 1 through 5 (early) while the other ANOVA had 15 levels of Block, corresponding to blocks 8 through 22 (late). Again, no analyses reached significance. For brevity, we report only the main effects of Boundary Orientation in each space (Blocks 1-5: factorial space: $F(1,30) = 1.5$, $MSe = .065$, $p = .23$, $\eta^2 = .048$; blended space: $F(1,30) < 1$; Blocks 8-22: factorial space: $F(1,30) < 1$; blended space: $F(1,30) = 1.6$, $MSe = .111$, $p = .21$, $\eta^2 = .051$).

Even though all of these analyses suggest that the spaces have little dimensional structure prior to category learning, we acknowledge that these null results could be due to a lack of power. The most important point to take away from these analyses is not the complete absence of dimensional structure in either space, but rather that the spaces do not differ in the degree to which they might possess dimensional structure. Thus differences in a priori dimensional structure cannot account for the results of Experiment 1.

^v Like Gureckis and Goldstone, we also observed increases in discriminability overall for pairs that crossed the middle of the space in both the relevant and irrelevant directions.

^{vi} Our findings could be relevant to other theoretical distinctions as well, such as the distinction between rule based and information integration based category learning. For instance, one might speculate that differentiation of the factorial space allows a kind of rule based categorization to take place, which in turn leads to perceptual learning specific to the relevant dimension (Ashby, Alfonso-Reese, Turken, & Waldron, 1998). The blended space, apparently resistant to differentiation, might allow only information integration. Of course single-system perspectives can also account for this phenomenon (Palmeri & Gauthier, 2004). Differentiation of the

factorial space could allow the selective attention parameter to weight the diagnostic dimension more heavily, stretching psychological space and resulting in further perceptual learning along the relevant dimension. Failure to differentiate the blended space could result in equal weighting for all features.

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FIGURE CAPTIONS

Figure 1. The top panel illustrates the factorial space used by Gurekis and Goldstone (2008) and Goldstone and Steyvers (2001). Panel a) shows an adaptation of the figure that appeared in Goldstone and Steyvers (2001). Panel b) shows a more abstract version of the space. Panel c) shows that space positioned within the tetrahedral volume defined by all possible morphs between four parents. The bottom panel illustrates the blended object space used by Jiang et al. (2007). Panel d) shows the figure that appeared in that paper. Panel e) shows a more abstract version of the space with parents at each of the corners. Panel f) shows that space more accurately as the full tetrahedral volume.

Figure 2. Factorial and blended spaces used in Experiment 1. Panels a) and d) show the spaces positioned within the full tetrahedron. The black dots at the grid intersections show the positions of the objects used in the category learning task and the open squares show the positions of the objects used in the discrimination task. Panels b) and e) show the relative positions of the categorization and discrimination objects (closed circles and open squares respectively). In b) the axes correspond to morphlines between parent cars. Points in the space are blends between particular positions along each axis. In e) all of the edges of the space correspond to morphlines between parent cars and blending is not factorial – rather each morph is created based on its relative proximity to each of the four corners. As a result, e) does not include axes. Panels c) and f) show samples from two spaces. The positions of the sample images were selected to maximize visual discriminability and the ability to discern the properties of the two spaces, and do not correspond to the actual stimuli used in the experiment, which were somewhat more compressed within the respective spaces.

Figure 3. Discrimination performance before and after training for object pairs along the relevant dimension vs. irrelevant dimension. The left panel shows the factorial space and the right panel shows the blended space. Error bars in this and all figures are between-subjects standard errors.

Figure 4. Each bar shows the difference between discrimination performance before and after training. Object pairs that cross the middle of the space are compared to pairs that do not cross the middle of the space in the relevant and irrelevant directions. A classical categorical perception effect predicts a greater training effect for middle pairs than outer pairs in the relevant but not the irrelevant directions. In fact, training appears to affect middle pairs slightly more than outer pairs in all directions in both spaces. Error bars are between-subjects standard errors.

Figure 5. Illustration of the design of Experiment 2. Dots show positions of the objects within the morphspaces in Experiment 2 (see also Figure 6). See main text for explanations of each panel.

Figure 6. Positions of the objects in Experiment 2, shown superimposed on the factorial and blended morphspaces within the tetrahedron (panels a and c) and schematically on flat surfaces.

Figure 7. Learning curves for first training phase for orthogonal (solid line) and diagonal (dashed line) category boundaries. Error bars are between-subjects standard errors.

Figure 8. More exemplars change categories in the 90 degree rotation than 45 degree rotation condition. Exemplars that change categories are shown as darker dots.

Figure 9. Learning curves for second training phase after 90 degree (solid line) and 45 degree (dashed line) rotations relative to category boundary learned during the first training phase. Error bars are between-subjects standard errors.

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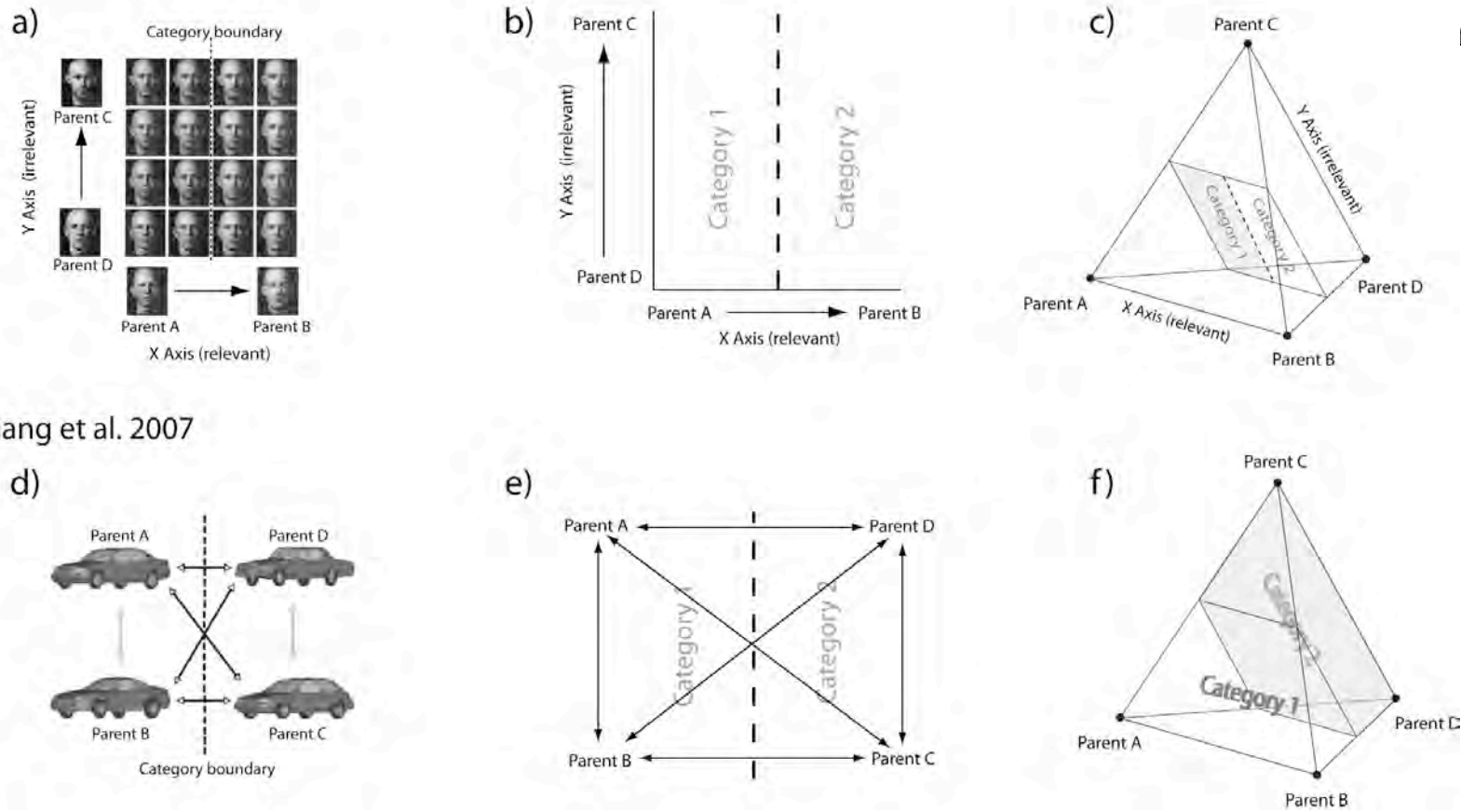
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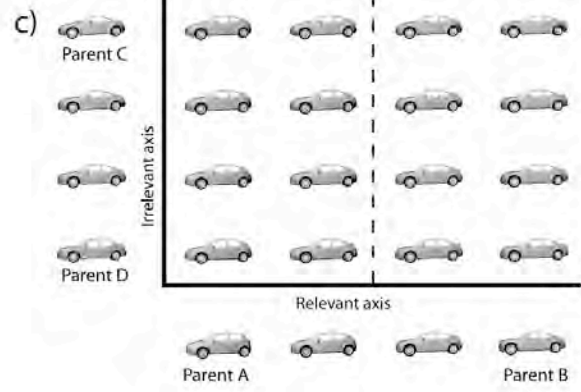
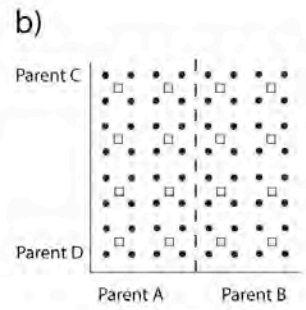
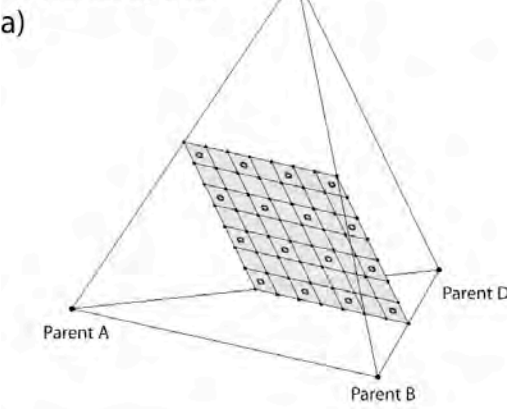
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Figure 1

Factorial space



Affects Object Representations

Blended space

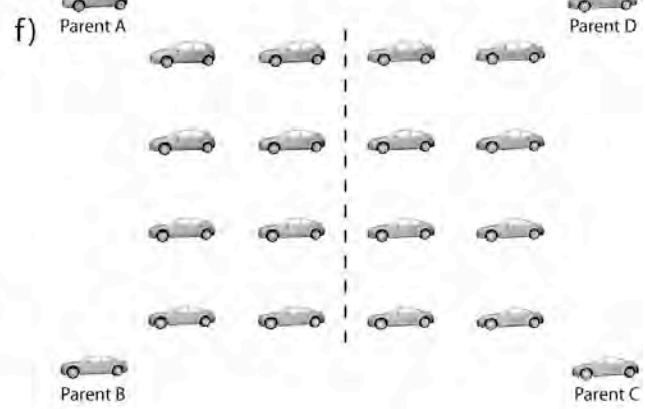
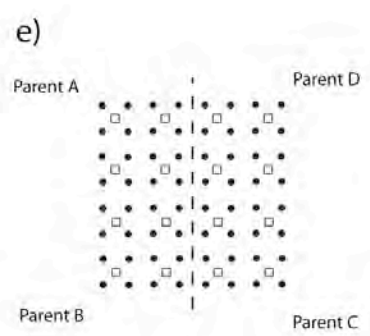
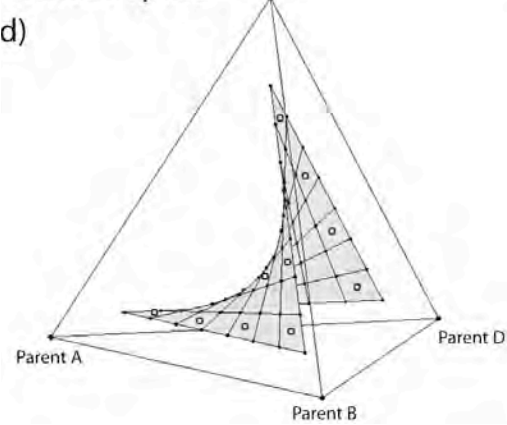


Figure 2

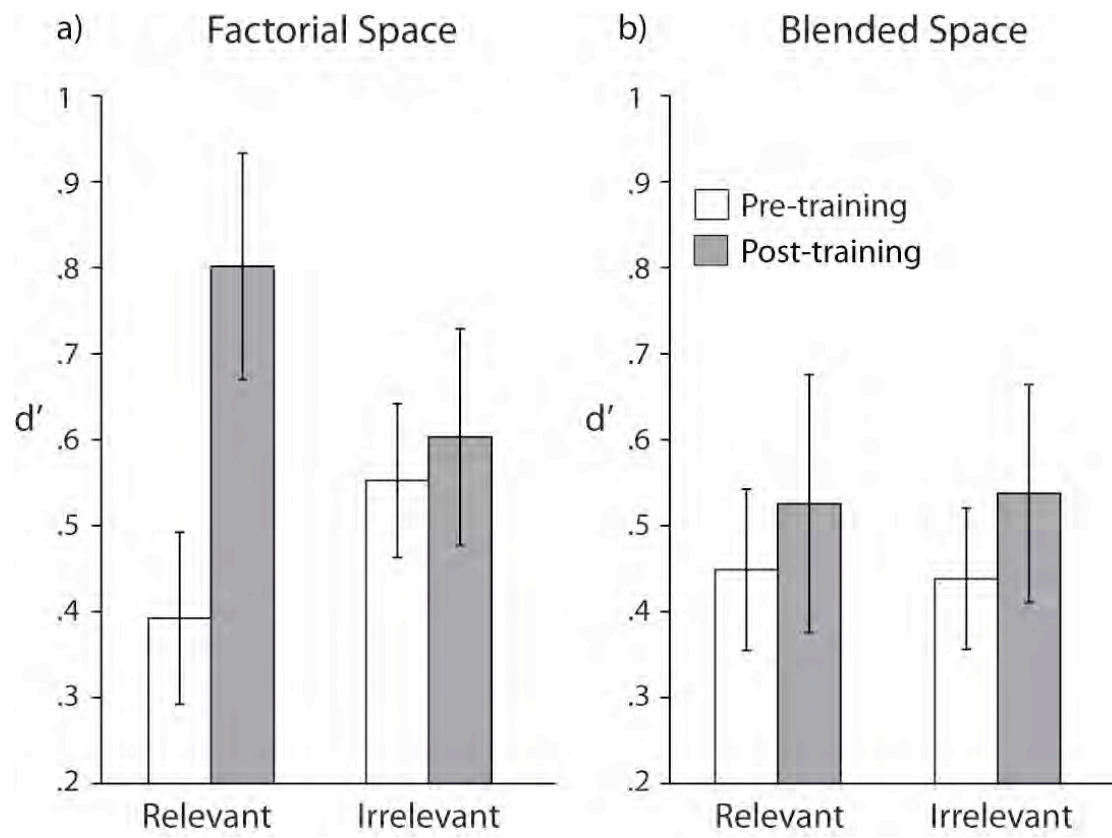


Figure 3

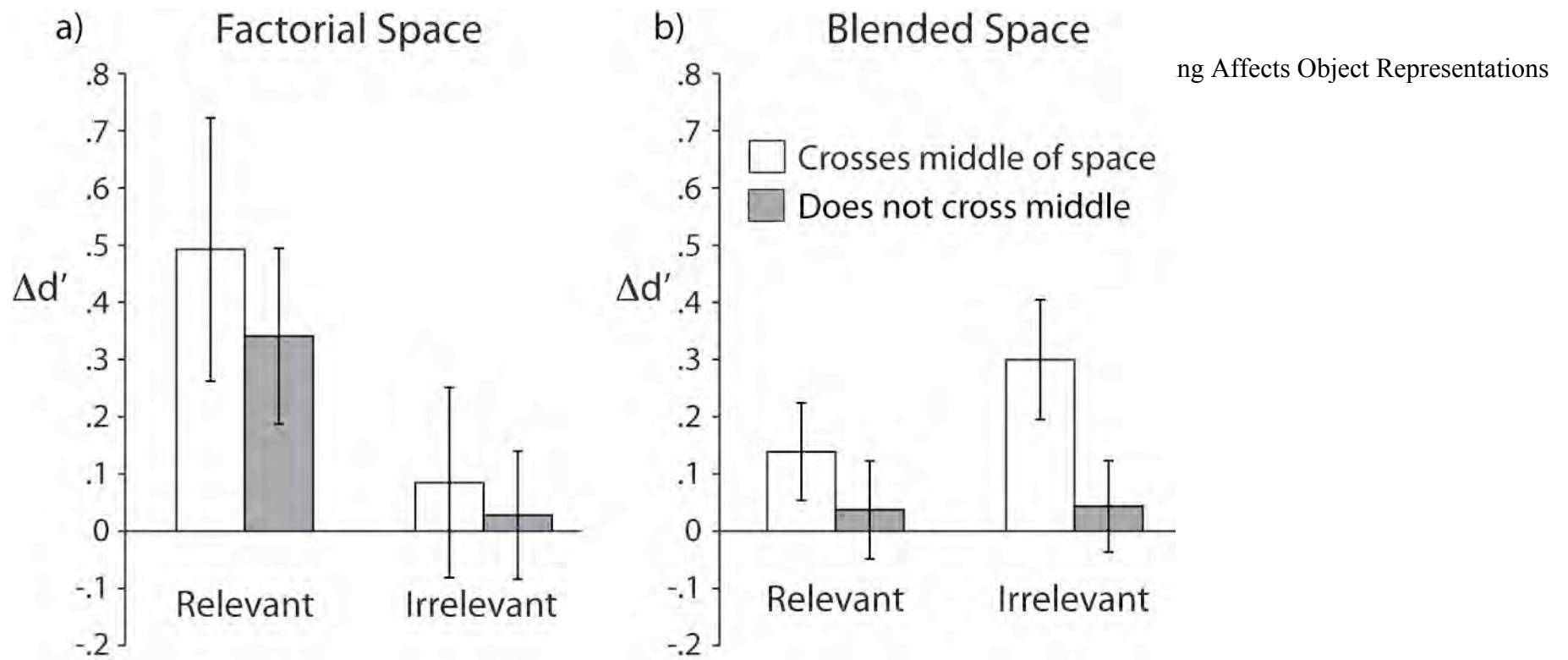


Figure 4

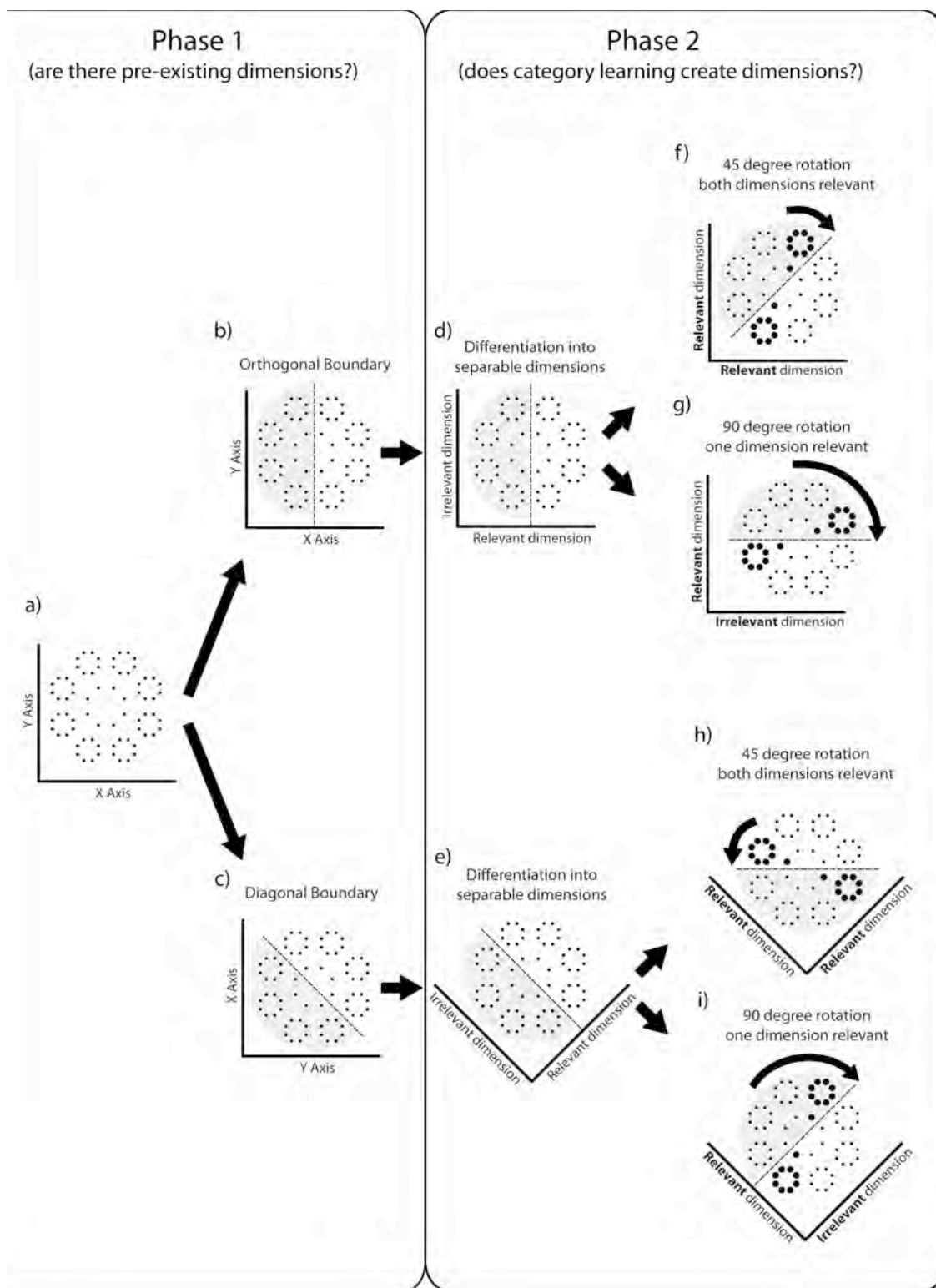


Figure 5

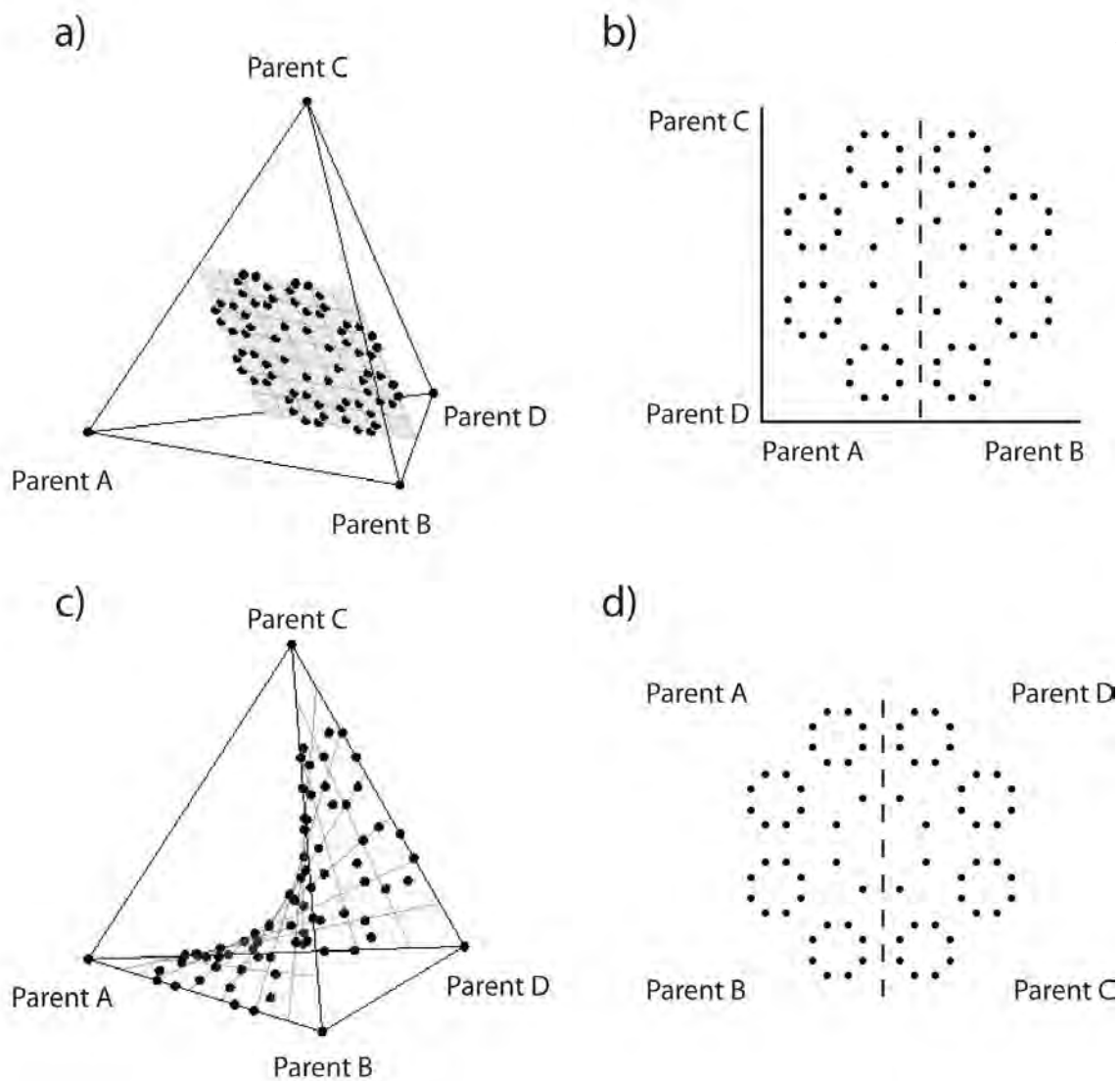


Figure 6

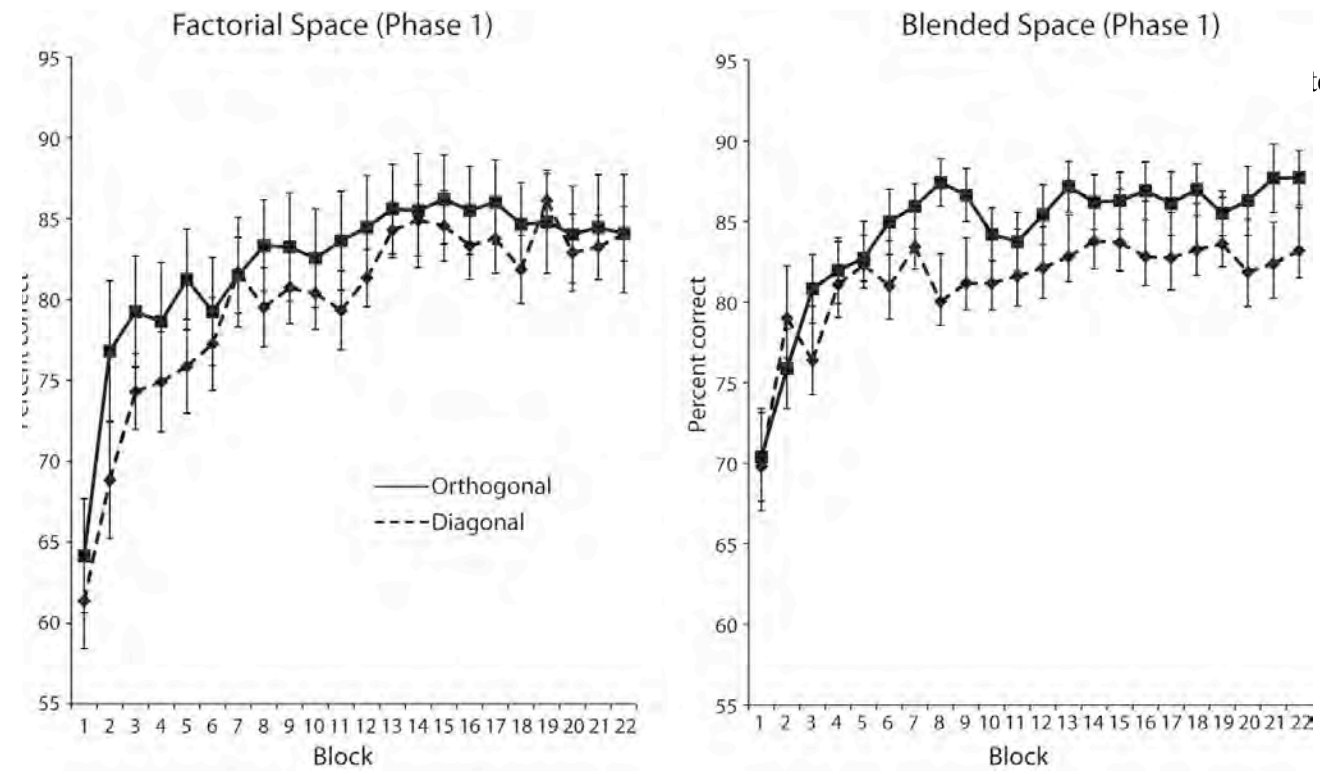
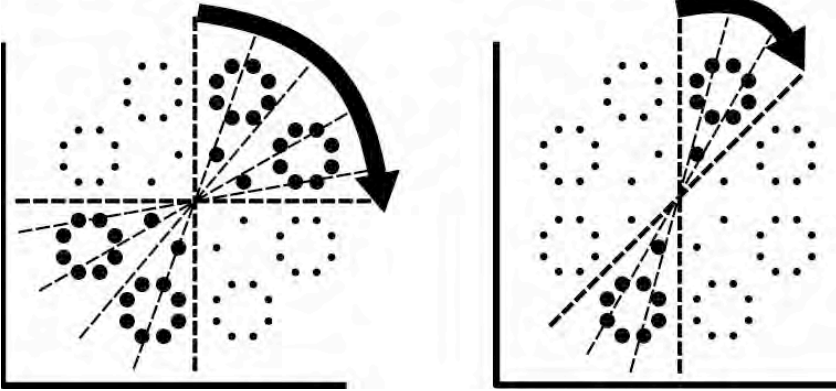


Figure 7.

30 degree rotation

45 degree rotation



How Category Learning Affects Object Representations

Figure 8

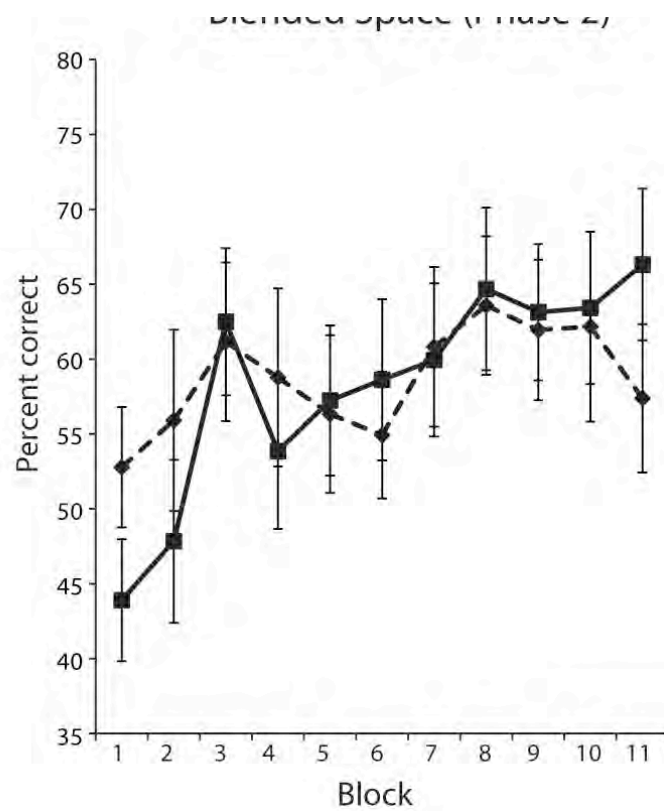
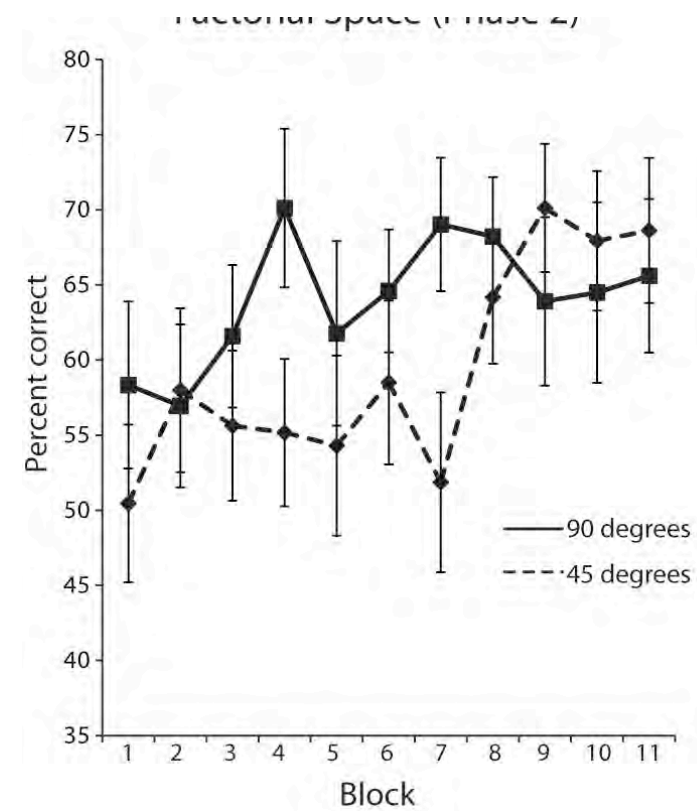


Figure 9